Semantic Communication for Edge AI

Deep joint source–channel coding with neural representation learning

This AI/ML project trains a neural encoder and dual-head decoder end to end through a differentiable noisy channel. Image classification is the evaluation task.

Capstone First Review

Review window: 18–22 August 2026

Presentation slot: 15 minutes

Team and guide details to be added

Edge AI inference over a noisy link

1

Observe

An edge camera captures an image.

2

Encode

A neural model learns what to transmit.

3

Communicate

The representation crosses a limited noisy link.

4

Infer

The receiver AI predicts the image class.

AI/ML core

Learn a compact, channel-robust representation that preserves the information needed for downstream inference.

Classification is the measurable AI task; neural representation learning is the method.

Conventional communication around an AI model

Image — camera input → **Compress** — JPEG 2000 → **Protect bits** — LDPC + modulation → **Add noise** — simulated channel → **AI classifier** — recovered image → ResNet-18

Compression	reduces the number of source bits.
Error correction	adds controlled redundancy so damaged bits can be recovered.
Modulation	maps protected bits to symbols sent through the channel.
AI classifier	a frozen ResNet-18 maps the recovered image to class logits.

Limitation: the communication stack is designed separately from the AI objective, even though task accuracy is the final metric.

The AI/ML method: end-to-end neural communication

Image	→	Residual CNN	→	AWGN layer	→	Dual-head decoder	→	Logits
sender input		learned representation		differentiable noise		class + reconstruction		AI output

The central change

Supervised backpropagation trains the sender and receiver together through the channel, so the transmitted representation can become robust to noise and useful for inference.

Why not send the label?

The project fixes the deployment split: the sender has an encoder, while the receiver owns the decoder and classifier. Sending a label would move the full task model to the sender and test a different system.

Training objective: cross-entropy(class) + λ × MSE(reconstruction)

PyTorch · residual CNN · GroupNorm/PReLU · supervised end-to-end learning

AI/ML and communication research gap

Short-packet communication	Finite packets pay coding overhead and can fail abruptly.	Measure the practical baseline instead of assuming ideal capacity. [1–4]
Learned image compression	Neural transforms learn compact image representations.	Good reconstruction quality is not the same as good AI task accuracy. [11–17]
DeepJSCC	Neural encoders and decoders can learn continuous channel mappings.	Most image studies emphasize reconstruction rather than classification. [5–10,24]
Task-oriented edge AI	Learned features can preserve inference under link limits.	Representation learning and joint coding are often not isolated. [18–23]

Gap addressed

A controlled three-way experiment that separates neural representation learning from joint channel coding at equal channel use.

AI/ML problem statement and objectives

Problem statement

A receiver-side AI model needs task-relevant information from an edge sensor, but a conventional link is optimized separately for bit and image recovery. The project tests whether a neural communication model can learn a better representation for inference.

Research question

At the same bandwidth and channel conditions, can a neural DJSCC model learn channel-robust representations that preserve AI inference accuracy better than tuned digital controls?

Objectives

1. Implement and train a residual neural encoder with a dual-head decoder.
2. Build a task-aware digital feature control using the shared AI representation.
3. Match channel use, SNR, image identity and noise across all systems.
4. Freeze all learned models and choices, then run one paired test campaign.

Completion depends on executing the fixed protocol correctly, not on obtaining a favorable result.

AI/ML architecture and comparison design

Imagenette-160 · same image and split · same complex-symbol budget · same SNR · same keyed noise

System	Transmission path	Scientific role
Classical + AI	JPEG 2000 → LDPC → AWGN → recovered image → frozen ResNet-18	Conventional reference
AI feature control	Shared neural encoder → quantization → same digital PHY → shared task head	Isolates representation learning
Neural DJSCC	Residual encoder → normalization → AWGN → dual-head neural decoder	Tests end-to-end neural coding

Evaluation controls Train with supervised CE + λMSE; tune on validation only; freeze before test access; retain failed transmissions; compare paired per-image AI outcomes.

The test split remains sealed until gate G-12. No test result exists at First Review.

AI evaluation hypotheses and analysis plan

H1 — low-SNR separation (confirmatory primary)

Comparator: learned versus classical_adaptive at the headline ratio. A grid point qualifies when the studentized paired mean $\sqrt{N}\cdot\hat{\Delta}(s)/\hat{\sigma}(s)$ exceeds 1.96. Let R_obs be the longest run of consecutive qualifying points at or below the training SNR. H1 is supported only if $R_{obs} \geq 3$ and the calibrated run p-value ≤ 0.05.

The whole low-SNR-region mean paired difference is the effect size of record. A curve crossing is reported if observed but is not required.

H2	Does a fixed conventional configuration show a cliff while the learned system degrades more gradually?
H3	Does the learned–classical accuracy gap contract as SNR improves?
H4	Does the learned joint system also exceed the task-aware digital feature control?

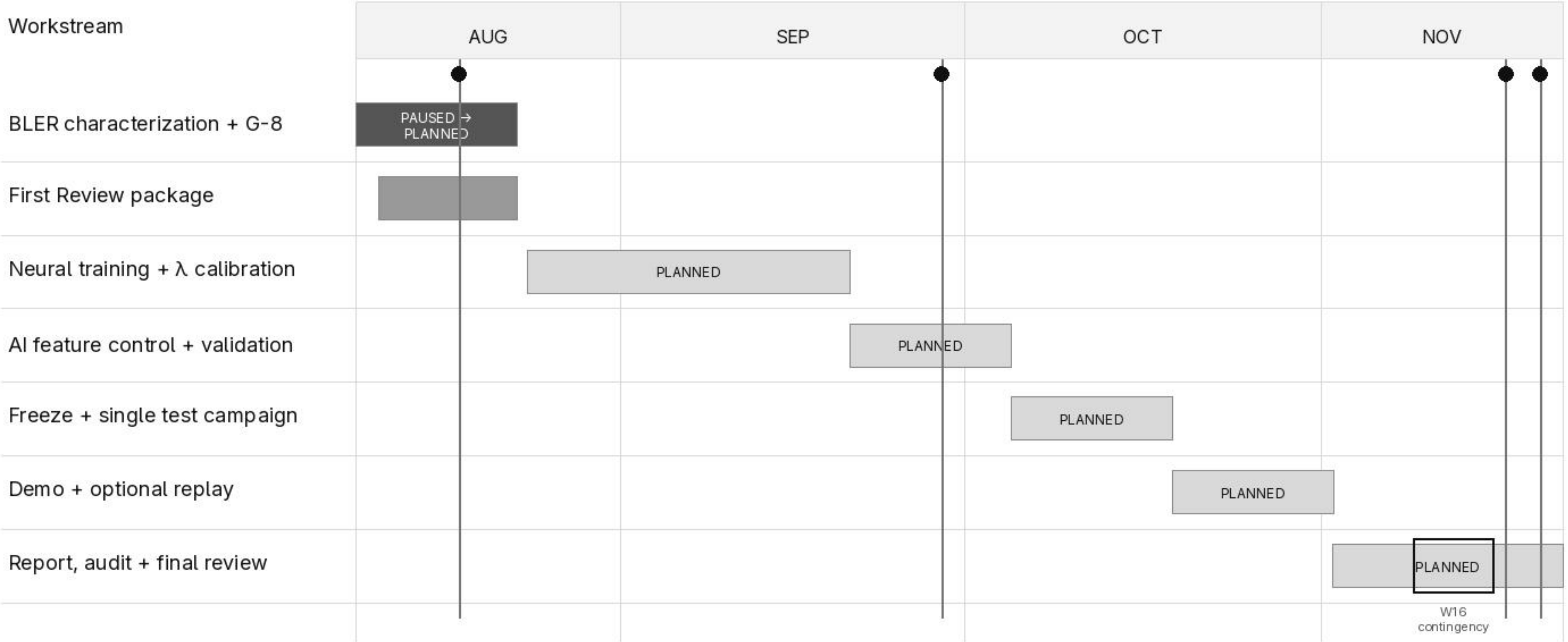
Outcome rule: support, no support or an adverse result are all valid findings. The protocol does not change after test access.

AI/ML feasibility demonstrated so far

Gate	Observed evidence	Plain-language meaning	Boundary
G-1	ResNet-18 from scratch: 89.8% validation	The AI task and clean reference model are viable.	Validation only; test is sealed.
G-7	Residual DJSCC: 1.64 M parameters	The neural model fits the RTX 4060 GPU profile.	A feasibility profile, not training.
G-2	Golden vectors and BLER conformance pass	The digital physical-layer implementation behaves as expected.	Small conformance evidence, not the full table.
W4	Bounded conventional pipeline runs end to end	Compression, coding, channel, decoding and records are integrated.	Integration evidence, not a comparison result.
G8_C	C2 characterization paused at a durable checkpoint	Full-strength BLER measurement can resume safely.	Table, selection and training are unfinished.

No worker is running. No selection, training, validation decoding or test access has occurred.

Project Gantt chart



Review dates: 18–22 Aug · 29 Sep–3 Oct · 17–21 Nov · report due 20 Nov

Dark bars show current Review 1 work; light bars are planned. W16 is report contingency, not new experiment scope.

Edge AI application, AIML contribution and risks

Edge AI application

Remote cameras and edge sensors send task-relevant information to a receiver-side AI model when bandwidth, latency or link reliability is limited. This represents split edge/cloud inference.

AI/ML contribution

Channel-robust neural representation learning; a residual encoder and dual-head decoder; CE + λMSE multi-task training; and a digital feature control that isolates the value of joint coding.

Main risks and controls

BLER work is incomplete: resume from the authenticated checkpoint.

A hypothesis may not be supported: report the result without retuning.

Hardware adds confounders: keep it as gated stretch work.

Technical claim boundary Tier 1 is a simulation-based AI experiment over normalized AWGN. OpenJPEG 2.5.4 and TS 38.212-derived LDPC are controls; no complete 5G NR radio or required hardware is claimed.

Outstanding human item: the guide's dated acknowledgement of the no-required-hardware path remains pending.

Summary and next steps

Project summary

Problem	Edge AI inference must operate over a limited noisy link.
AI/ML method	Train a residual neural encoder and dual-head decoder through a differentiable channel.
Evaluation	Measure Imagenette-160 classification at matched channel use against two controlled baselines.
Next work	Finish characterization, train and freeze the models, then run the paired test campaign once.

Six-criterion rubric map

Motivation	2–4, 6, 11
Objectives	6–7, 11–12
Hypothesis	8
Problem Survey	5
Subject Knowledge	3–5, 7–9, 11
Time Plan	10

First Review position The AI/ML method, controlled evaluation protocol, feasibility evidence and completion plan are defined. Final neural training and learned-versus-classical results remain future work.